COMPANION RUNBOOK / WINDOWS ADMINISTRATION

Windows Server Troubleshooting Command Guide

Oracle Database 19c | Windows Server 2025 Standard Evaluation

Beginner-friendly PowerShell and Windows GUI reference. Built around the verified Oracle lab with ORCLCDB, ORCLPDB1 and TCP listener 1521.

Guide details	Value
Edition / date	v1.0	03 November 2026
Target environment	WIN-JE797ASF0CA / Windows Server 2025 Standard Evaluation (build 26100)
Oracle Home	C:\app\oracle\product\19.0.0\dbhome_1
Oracle Base	C:\app\oracle
Database services	ORCLCDB (CDB), ORCLPDB1 (PDB), LISTENER TCP 1521
Audience	Windows Server beginners supporting an Oracle Database lab
Command safety	Read-only by default; changes are explicitly called out

SUPPORT / SECURITY: Oracle Database 19.3 is not certified for Windows Server 2025. The minimum supported 19c RU is 19.27. Keep this unpatched configuration isolated and non-production; these commands do not remedy that support limitation.

At a glance: first response to an incident

1. Identify the Windows host, OS version, current account and recent reboot time.
2. Check Oracle database and listener service state.
3. Verify listening TCP port 1521, DNS resolution and client-to-listener reachability.
4. Use SQL*Plus and listener diagnostics to separate instance failure from Oracle Net failure.
5. Read Windows event logs and Oracle diagnostic logs; capture exact timestamps and error codes.
6. Only after the fault is identified should a controlled, documented configuration change be considered.

IMPORTANT PROMPT DISTINCTION: Commands beginning Get-, Test-, or Get-Content run in PowerShell (PS>). Commands such as SHOW PDBS or SELECT ... FROM v$database run inside SQL*Plus (SQL>). lsnrctl and tnsping are executables launched from PowerShell.

1. Windows command environments

1.1 Where to type which command

Prompt / application	Use for	Example
PowerShell (PS>)	Windows OS, services, files, ports, logs	Get-Service *Oracle*
Command Prompt (C:\>)	Legacy utilities; .exe commands	ipconfig /all
SQL*Plus (SQL>)	Oracle SQL and SQL*Plus commands	SHOW PDBS;
RMAN (RMAN>)	Oracle backup/recovery commands	LIST BACKUP SUMMARY;

In PowerShell, the ampersand (&) runs an executable whose path is quoted. Without &, a quoted executable path is only a string.

POWERSHELL | COPY-READY COMMANDS

```
& "C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe" -v
Get-Command Get-NetTCPConnection -Syntax
Get-Help Get-Service -Examples
```

1.2 Linux-to-PowerShell vocabulary

Linux habit	Windows / PowerShell equivalent	Meaning
pwd	Get-Location	Current directory
ls -la	Get-ChildItem -Force	Show files, including hidden
cat file	Get-Content file	Read file
tail -n 50 file	Get-Content file -Tail 50	Last 50 lines
tail -f file	Get-Content file -Wait -Tail 20	Follow log (Ctrl+C to exit)
grep ERROR file	Select-String -Path file -Pattern ERROR	Search text
which sqlplus	Get-Command sqlplus.exe	Find executable in PATH
ps aux	Get-Process	Processes
systemctl status x	Get-Service x	Service state
ss -lntp	Get-NetTCPConnection -State Listen	Listening TCP ports
df -h	Get-Volume	Volume capacity

1.3 Useful Windows graphical consoles

Launch from Win+R or PowerShell	Purpose
services.msc	Manage Windows services
eventvwr.msc	View Windows event logs
taskmgr.exe	Task Manager
resmon.exe	Resource Monitor
perfmon.exe	Performance Monitor
diskmgmt.msc	Disk Management
wf.msc	Windows Defender Firewall with Advanced Security
ncpa.cpl	Network Connections
compmgmt.msc	Computer Management

2. OS identity, user and reboot history

POWERSHELL | COPY-READY COMMANDS

```
Get-CimInstance Win32_OperatingSystem |
  Select-Object Caption,Version,BuildNumber,OSArchitecture,LastBootUpTime
$env:COMPUTERNAME
whoami
(Get-Date) - (Get-CimInstance Win32_OperatingSystem).LastBootUpTime
```

Interpretation: Captures OS edition/build, hostname, the Windows account executing the commands, and uptime. Use these details in any incident ticket.

2.1 Time and regional settings

POWERSHELL | COPY-READY COMMANDS

```
Get-Date
Get-TimeZone
w32tm /query /status
```

Your Oracle diagnostic logs showed a -08:00 offset while your working location may use a different time zone. Compare Windows time and log timestamps when correlating incidents. Do not change server time-zone settings without validating the application requirements.

3. Windows services and startup configuration

POWERSHELL | COPY-READY COMMANDS

```
Get-Service *Oracle*
Get-CimInstance Win32_Service |
  Where-Object { $_.Name -match 'Oracle' } |
  Select-Object Name,State,StartMode,StartName
sc.exe qc OracleServiceORCLCDB
Get-Service -Name OracleServiceORCLCDB,OracleOraDB19Home1TNSListener
```

Expected in your lab: OracleServiceORCLCDB and OracleOraDB19Home1TNSListener are Running with Automatic startup. OracleJobSchedulerORCLCDB may remain Disabled. OracleVssWriterORCLCDB may remain Stopped when VSS backups are not running.

CAUTION: Get-Service and sc.exe qc are read-only. Stopping or restarting OracleServiceORCLCDB disrupts database availability. Never use Stop-Service, Restart-Service, taskkill or reboot as a diagnostic shortcut while clients are connected.

3.1 Registry verification - read-only

POWERSHELL | COPY-READY COMMANDS

```
$key='HKLM:\SOFTWARE\ORACLE\KEY_OraDB19Home1'
Get-ItemProperty $key | Select-Object ORACLE_HOME,ORACLE_BASE,ORACLE_HOME_NAME
Get-ItemProperty $key |
  Select-Object ORA_ORCLCDB_AUTOSTART,ORA_ORCLCDB_SHUTDOWN,ORA_ORCLCDB_SHUTDOWNTYPE
```

Your verified values were AUTOSTART=TRUE, SHUTDOWN=TRUE and SHUTDOWNTYPE=immediate. The registry is the authoritative installation metadata for this Windows Oracle Home; avoid creating unnecessary machine-wide ORACLE_HOME variables.

4. IP addresses, routing and DNS

POWERSHELL | COPY-READY COMMANDS

```
Get-NetIPConfiguration
Get-NetAdapter | Select-Object Name,Status,LinkSpeed
Get-NetIPAddress -AddressFamily IPv4 |
 Select-Object InterfaceAlias,IPAddress,PrefixLength
Get-NetRoute -AddressFamily IPv4 |
 Where-Object DestinationPrefix -eq '0.0.0.0/0'
Resolve-DnsName WIN-JE797ASF0CA
ipconfig /all
```

If the hostname resolves to multiple addresses, compare the results with the address the Oracle client actually uses. On your VM, the host previously resolved to both an IPv6 link-local address and 192.168.122.120.

4.1 Test hostname and port reachability

POWERSHELL | COPY-READY COMMANDS

```
Test-NetConnection 127.0.0.1 -Port 1521
Test-NetConnection WIN-JE797ASF0CA -Port 1521
Test-NetConnection 192.168.122.120 -Port 1521
Test-NetConnection 127.0.0.1 -Port 5500
```

TcpTestSucceeded=True proves a TCP connection could be established; it does NOT prove Oracle login, service resolution, or TLS/application-layer behavior. Port 5500 is the EM Express TCPS listener port in this lab.

4.2 Find the process listening on a port

POWERSHELL | COPY-READY COMMANDS

```
Get-NetTCPConnection -LocalPort 1521 -State Listen |
 Select-Object LocalAddress,LocalPort,OwningProcess
Get-Process -Id 8124 -ErrorAction SilentlyContinue
netstat -ano | findstr :1521
```

8124 was the listener process ID during one earlier capture; it may change after reboot. Use the current OwningProcess from the first command instead of assuming a fixed PID. LocalAddress :: means an IPv6 wildcard listener, not necessarily a failure.

5. Windows Firewall - inspect without disabling

POWERSHELL | COPY-READY COMMANDS

```
Get-NetFirewallProfile |
 Select-Object Name,Enabled,DefaultInboundAction,DefaultOutboundAction
Get-NetFirewallPortFilter -Protocol TCP |
 Where-Object { $_.LocalPort -eq '1521' } |
 Get-NetFirewallRule |
 Select-Object DisplayName,Enabled,Direction,Action
Get-NetConnectionProfile
```

Review the complete firewall rule scope and active network profile. A rule existing for TCP 1521 does not necessarily mean all remote IP addresses are permitted. A local Test-NetConnection success also does not prove that a remote client can connect.

CAUTION: Do not disable the firewall, create wide-open inbound rules, or expose ports 1521 or 5500 to the public internet. If a rule is needed, require authorized source subnets, a change record and a rollback plan.

6. Files, disk space and NTFS permissions

POWERSHELL | COPY-READY COMMANDS

```
Get-Volume | Select-Object DriveLetter,FileSystemLabel,FileSystem,SizeRemaining,Size
Get-PSDrive -PSProvider FileSystem
Get-Disk | Select-Object Number,FriendlyName,OperationalStatus,Size
Test-Path 'C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe'
Get-ChildItem 'C:\app\oracle' -Force
icacls 'C:\app\oracle'
(Get-Acl 'C:\app\oracle').Access |
 Select-Object IdentityReference,FileSystemRights,AccessControlType
```

Common interpretation: Test-Path=False indicates a missing path; permission denied requires account/ACL analysis; low SizeRemaining can affect the FRA, archive logging, Oracle startup and backup operations. Avoid copying, deleting or modifying live database datafiles through File Explorer.

6.1 Temporary files and environment

POWERSHELL | COPY-READY COMMANDS

```
$env:TEMP
$env:TMP
Get-ChildItem Env: | Where-Object Name -Match 'ORACLE|TNS|PATH'
$env:PATH -split ';' | Where-Object { $_ -match 'oracle' }
```

Use full paths for Oracle executables if their bin folder is not on PATH. Environment variables defined with $env: in PowerShell normally apply only to the current process and child processes, not permanently.

7. Processes, RAM, CPU and performance

POWERSHELL | COPY-READY COMMANDS

```
Get-Process | Sort-Object CPU -Descending |
 Select-Object -First 12 Name,Id,CPU,@{N='WorkingSetMB';E={[math]::Round($_.WorkingSet64/1MB)}}
Get-CimInstance Win32_OperatingSystem |
 Select-Object @{N='TotalRAM_GB';E={[math]::Round($_.TotalVisibleMemorySize/1MB,2)}},
          @{N='FreeRAM_GB';E={[math]::Round($_.FreePhysicalMemory/1MB,2)}}
Get-Counter '\Processor(_Total)\% Processor Time' -SampleInterval 1 -MaxSamples 3
```

CPU from Get-Process is cumulative processor time, not current utilization percentage. Use performance counters or Resource Monitor to assess current CPU load. RAM figures from Win32_OperatingSystem are in KB and are converted above to GiB.

8. Windows Event Logs

POWERSHELL | COPY-READY COMMANDS

```
Get-WinEvent -LogName System -MaxEvents 20 |
 Select-Object TimeCreated,Id,LevelDisplayName,ProviderName,Message
Get-WinEvent -FilterHashtable @{
 LogName='Application'; StartTime=(Get-Date).AddHours(-24); Level=2,3
} -MaxEvents 30 |
 Select-Object TimeCreated,Id,ProviderName,Message
Get-WinEvent -FilterHashtable @{
 LogName='System'; StartTime=(Get-Date).AddHours(-24); Level=2,3
} -MaxEvents 30 |
 Select-Object TimeCreated,Id,ProviderName,Message
```

Level 2=Error and Level 3=Warning. Correlate Windows event times with the Oracle alert and listener logs; failures may be reported under Application, System, service control manager or Oracle-related providers.

8.1 Save a small event-log extract

POWERSHELL | COPY-READY COMMANDS

```
Get-WinEvent -LogName System -MaxEvents 100 |
 Select-Object TimeCreated,Id,ProviderName,LevelDisplayName,Message |
 Export-Csv "$env:USERPROFILE\Desktop\system-events.csv" -NoTypeInformation
```

The exported CSV is an evidence file; review it for hostnames, usernames and sensitive operational details before sharing it.

9. Oracle tools from Windows PowerShell

Use the actual installed Oracle Home, not a bare sqlplus/lsnrctl command, if PATH is unconfigured.

POWERSHELL | COPY-READY COMMANDS

```
$OH='C:\app\oracle\product\19.0.0\dbhome_1'
& "$OH\bin\sqlplus.exe" -v
& "$OH\bin\lsnrctl.exe" status LISTENER
& "$OH\bin\lsnrctl.exe" services LISTENER
& "$OH\bin\tnsping.exe" '127.0.0.1:1521/orclpdb1'
```

LSNRCTL status shows listening endpoints and registered services. LSNRCTL services shows handlers and counters. TNSPING checks naming and listener contact; it does not prove database login succeeds.

9.1 SQL*Plus local SYSDBA verification

POWERSHELL | COPY-READY COMMANDS

```
$env:ORACLE_SID='ORCLCDB'
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe' / as sysdba
```

SQL*PLUS | COPY-READY COMMANDS

```
SHOW CON_NAME;
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
SELECT instance_name,host_name,status FROM v$instance;
EXIT;
```

A local CONNECT / AS SYSDBA uses Windows OS authentication and can succeed even when TCP Oracle Net login is failing. Do not mistake that success for proof that remote clients can connect.

9.2 Password-authenticated PDB connectivity

POWERSHELL | COPY-READY COMMANDS

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe' /nolog
```

SQL*PLUS | COPY-READY COMMANDS

```
CONNECT SYSTEM@"127.0.0.1:1521/orclpdb1"
-- At the password prompt, enter the secret privately.
-- In this lab, enclosing a password containing @ in double quotes worked.
SHOW CON_NAME;
SHOW USER;
EXIT;
```

SOLVED IN THIS LAB: The earlier ORA-12543 was caused by SQL*Plus parsing a password containing @ as a destination. The client diagnostic file showed HOST=0.0.0.123 and SERVICE_NAME empty. Entering that password in double quotes at the password prompt restored successful SYSTEM login to ORCLPDB1. Never paste or log real passwords.

10. Read Oracle diagnostic logs

10.1 Listener log

POWERSHELL | COPY-READY COMMANDS

```
$listenerLog='C:\app\oracle\diag\tnslsnr\WIN-JE797ASF0CA\listener\trace\listener.log'
Test-Path $listenerLog
Get-Content $listenerLog -Tail 80
Select-String -Path $listenerLog -Pattern 'TNS-|ORA-|establish' |
  Select-Object -Last 20
```

10.2 SQL*Plus client sqlnet.log

POWERSHELL | COPY-READY COMMANDS

```
$clientLog=Get-ChildItem 'C:\app\oracle\diag\clients' `
  -Filter sqlnet.log -File -Recurse -ErrorAction SilentlyContinue |
  Sort-Object LastWriteTime -Descending | Select-Object -First 1
$clientLog.FullName
if ($clientLog) { Get-Content $clientLog.FullName -Tail 60 }
```

10.3 Database alert log

POWERSHELL | COPY-READY COMMANDS

```
Get-ChildItem 'C:\app\oracle\diag\rdbms' -Filter 'alert_*.log' `
  -Recurse -File -ErrorAction SilentlyContinue |
  Select-Object FullName,LastWriteTime
# Replace this value with the FullName returned above:
# Get-Content 'C:\app\oracle\diag\rdbms\...\trace\alert_orclcdb.log' -Tail 80
```

Read-only alternatives include Oracle ADRCI, but searching the diagnostic directory is usually simpler for a beginner. Capture the full error stack, timestamp and surrounding entries, not only the final ORA number.

11. Oracle database health queries

Run the following in SQL*Plus as a suitably privileged administrator. Confirm the container with SHOW CON_NAME first.

SQL*PLUS (CDB ROOT) | COPY-READY COMMANDS

```
SHOW CON_NAME;
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
SELECT con_name,state,restricted FROM dba_pdb_saved_states;
SHOW PARAMETER db_recovery_file_dest;
SHOW PARAMETER db_recovery_file_dest_size;
```

To inspect your APP_DATA tablespace, switch to the PDB:

SQL*PLUS (PDB) | COPY-READY COMMANDS

```
ALTER SESSION SET CONTAINER=ORCLPDB1;
SELECT tablespace_name,status,contents FROM dba_tablespaces;
SELECT tablespace_name,file_name,bytes/1024/1024 size_mb,
    autoextensible FROM dba_data_files
WHERE tablespace_name='APP_DATA';
```

The APP_DATA tablespace was actually created and verified at 100 MB initial size with autoextend to 1 GiB. The application user/sample tables, RMAN backup and Data Pump exercises were not yet executed when this guide was prepared.

11.1 Inspect recovery area utilization

SQL*PLUS | COPY-READY COMMANDS

```
ALTER SESSION SET CONTAINER=CDB$ROOT;
SELECT name,space_limit/1024/1024 limit_mb,
    space_used/1024/1024 used_mb,
    space_reclaimable/1024/1024 reclaimable_mb
FROM v$recovery_file_dest;
```

A full FRA can affect redo archiving and database availability. Do not manually delete archived redo logs to clear space; manage them using RMAN and an approved backup/retention policy.

12. RMAN and Data Pump command checks

These checks do not perform backups or exports. They are a safe way to verify installed utilities and inspect backup metadata.

POWERSHELL | COPY-READY COMMANDS

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\rman.exe' target /
```

RMAN | COPY-READY COMMANDS

```
SHOW ALL;
LIST BACKUP SUMMARY;
EXIT;
```

POWERSHELL | COPY-READY COMMANDS

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\expdp.exe' help=y
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\impdp.exe' help=y
```

RMAN LIST BACKUP SUMMARY may return no backups if the backup steps from the separate installation SOP have not been executed. Avoid entering Oracle passwords as command-line arguments; these can be captured in console history or process inspections.

13. Common symptoms and next checks

Symptom	Likely area	First checks
SQL*Plus ORA-12543 but TNSPING OK	Client descriptor or credential parsing	Client sqlnet.log; confirm HOST/SERVICE_NAME; check @ password quoting
ORA-12504	Service name omitted from request	Review connect string; check lsnrctl services
ORA-12541	No listener reachable	Get-Service, lsnrctl status, Test-NetConnection 1521
ORA-01017	Authentication or wrong target PDB	Check user, quoted password, SHOW CON_NAME
CDB works but PDB not open	PDB open state	SHOW PDBS; DBA_PDB_SAVED_STATES
Unable to create a datafile	Disk, path, ACL or storage parameter	Get-Volume, Test-Path, icacls, Oracle alert log
Database not started after reboot	Service/registry/PDB state	Get-Service, ORA_ORCLCDB_AUTOSTART, saved PDB state
FRA nearly full	Recovery logs/backup retention	V$RECOVERY_FILE_DEST and RMAN policy
EM Express does not open	TCPS endpoint, cert, port, firewall	lsnrctl status (5500), Test-NetConnection 5500

14. Minimal incident evidence bundle

For a new troubleshooting request, capture these outputs first; they are typically sufficient to identify the next diagnostic action without disruptive changes.

POWERSHELL | COPY-READY COMMANDS

```
$OH='C:\app\oracle\product\19.0.0\dbhome_1'
Get-CimInstance Win32_OperatingSystem |
 Select-Object Caption,Version,BuildNumber,LastBootUpTime
Get-CimInstance Win32_Service | Where-Object Name -match 'Oracle' |
 Select-Object Name,State,StartMode
Get-NetIPConfiguration
Test-NetConnection 127.0.0.1 -Port 1521
Get-NetTCPConnection -LocalPort 1521 -State Listen
& "$OH\bin\lsnrctl.exe" status LISTENER
& "$OH\bin\lsnrctl.exe" services LISTENER
```

For database-specific incidents, additionally capture SHOW CON_NAME, SHOW PDBS, the relevant SQL error, and the last 40-80 lines of the corresponding diagnostic log. Remove passwords and other secrets before sharing.

15. Read-only versus change commands

Category	Examples	Guidance
Read-only diagnostics	Get-Service; Get-WinEvent; Test-NetConnection; lsnrctl status; SELECT	Safe starting point; some logs and registry details require Administrator
Limited change	ALTER SESSION SET CONTAINER; setting $env:ORACLE_SID in current terminal	Affects the current session; still confirm context
Service interruption	Stop-Service; Restart-Service; SHUTDOWN IMMEDIATE; reboot	Requires window, approval and impact assessment
High-risk system change	Firewall rule edits; network reconfiguration; ACL edits; registry writes	Require backup/rollback and change control
Data destructive	Deleting Oracle datafiles; DROP DATABASE; deleting FRA files	Do not perform as a troubleshooting shortcut

16. Official references

Test-NetConnection: https://learn.microsoft.com/powershell/module/nettcpip/test-netconnection?view=windowsserver2025-ps

Get-NetTCPConnection: https://learn.microsoft.com/powershell/module/nettcpip/get-nettcpconnection?view=windowsserver2025-ps

Get-Service: https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.management/get-service

Get-WinEvent: https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.diagnostics/get-winevent

Get-NetFirewallPortFilter: https://learn.microsoft.com/powershell/module/netsecurity/get-netfirewallportfilter

Oracle automatic startup/shutdown on Windows: https://docs.oracle.com/en/database/oracle/oracle-database/19/ntqrf/automatic-startup-and-shutdown.html

Oracle 19c Windows installation guide: https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/

End of guide. This is a diagnostic companion to Oracle_19c_Windows_Server_2025_Installation_Admin_SOP_v1.0.